VrikkaTwin · one day in the life of the twin

Every morning the twin re-learns the patient and looks 72 hours ahead

Danger signs are checked first, at any time of day

EVERY MORNING AUTOMATIC

1 Readings — 08:00

- **Weight** (smart scale)
- **Resting heart rate** (band)
- No device: ASHA weekly visit

2 Integrity check

- Plausible value?
- Fresh, not replayed?
- Signed by the device?

3 Twin update

- Kalman filter re-estimates
- **body water** + **creatinine**
- Lab result used if one came in

4 Simulate 72 h — 09:00

- **200 possible futures**
- on the noisy weather forecast
- with today's tablets & symptoms

5 Model M

- LightGBM + 25 features
- Calibrated **% risk**
- **Top-3 reasons**

6 Alert level

✅ Green ⚠️ Amber ⛔ Red

- Red kept ≤ 0.5 per 100 days
- No alert fatigue

WHAT HAPPENS NEXT

✅ Green · low risk

- Carry on as usual
- Water through the day
- Shade when it's hot

⚠️ Amber · building

- ORS, avoid painkillers, shade
- 🔔 **Patient + family**
- 4 languages + audio

⛔ Red · high risk

- 🔔 **Patient, family, doctor**
- Doctor reviews the reasons
- Painkiller / medicine review

💉 Home kidney test

- Doctor orders in one click
- Sample collected at home
- About **₹189**

🔬 Lab result

- Entered by the lab
- Flows **back into the twin**
- ↻ tomorrow's step 3

🟣 Confirmed injury

- Lab-confirmed kidney injury
- Doctor's plan today
- Usually mild, recovers

ANY TIME CHECK-IN

💬 Check-in

- Patient, family or ASHA
- "bukhar hai, Combiflam li"
- Tablets + symptoms, any language

🚨 Red flags first

- Runs **before any AI**
- Confusion, fits, fainting
- ≥ 40 °C, no urine 12 h

🚑 Emergency

- **Call 108 / 112** button
- Shade, cool the body
- 🔔 Doctor + family at once

⚠️ Urgent

- See a doctor today
- Alert at least **red**
- Kidney blood test

✔ No danger sign

- Symptoms + tablets recorded
- Used in tonight's twin run
- 💊 Triple-whammy check

📒 Audit log

- Every step recorded
- Consent checked
- Tamper-evident

EXAMPLE (FICTIONAL)

👵 Riya, 68 · Nagpur
BP pill + water tablet → 🌡️ 45 °C heatwave
✅ green → ⚠️ amber → 🤒 Fever + Combiflam
⛔ **red 43 %** → ⏰ Warned **23 h** before
the simulated injury → 🔁 Swap painkiller + ORS
risk **48 % → < 2 %**

Synthetic & fictional patients only · not a medical device